

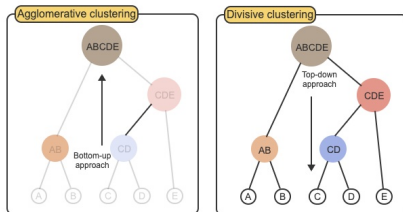
Unsupervised Learning

Hierarchical Clustering

Introduction to Hierarchical Clustering

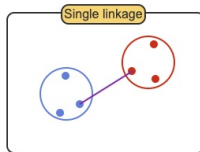
Hierarchical clustering groups instances into a hierarchy of clusters: at one end $\Rightarrow$ a single cluster with all instances; at the other end $\Rightarrow$ clusters with just one instance.

- **Agglomerative hierarchical clustering:** each instance is treated as an individual cluster; two clusters are combined iteratively until all instances belong to a single cluster (*bottom-up approach*).
- **Divisive hierarchical clustering:** all instances belong to a single cluster; each cluster is split into two clusters until all clusters contain just a single instance (*top-down approach*).

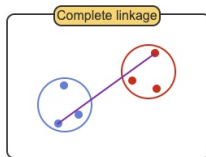


Measures of cluster similarity

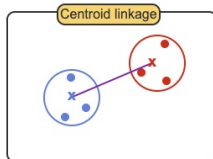
- ▶ Single linkage



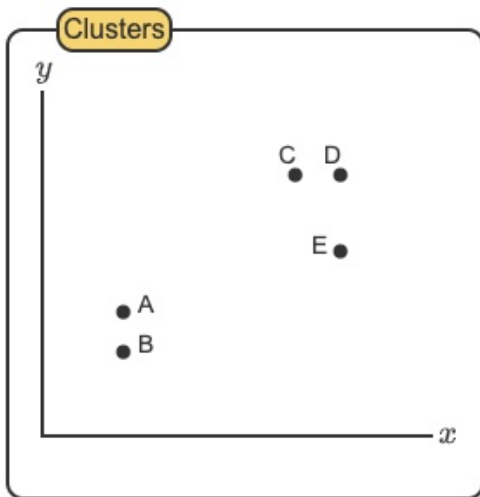
- ▶ Complete linkage



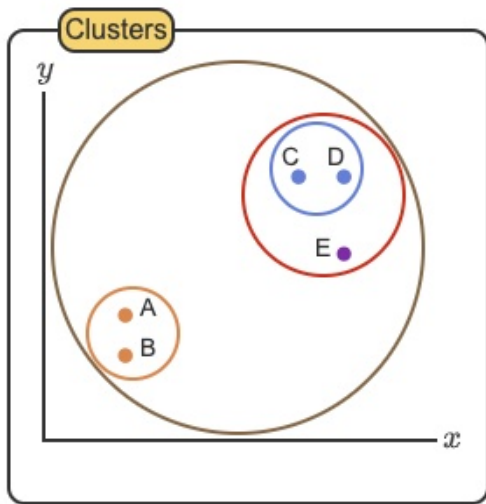
- ▶ Centroid linkage



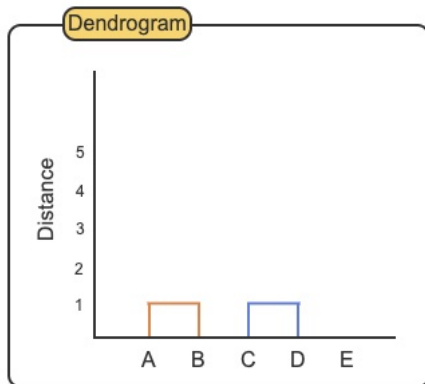
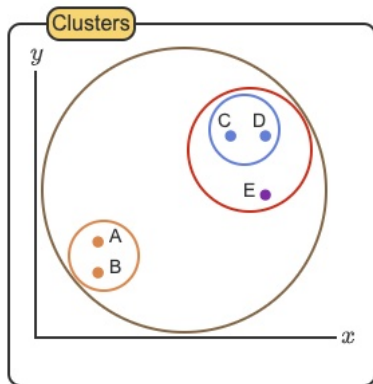
Dendrograms



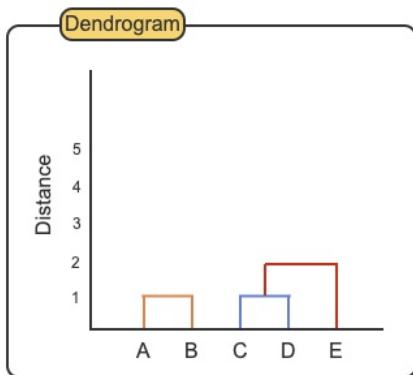
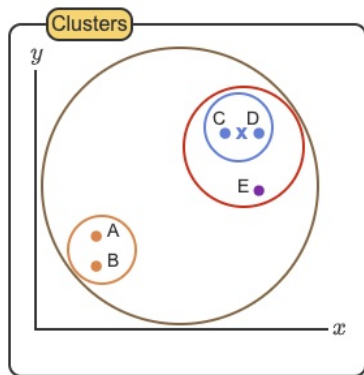
Dendrograms



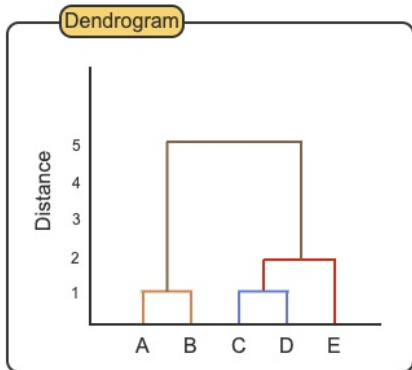
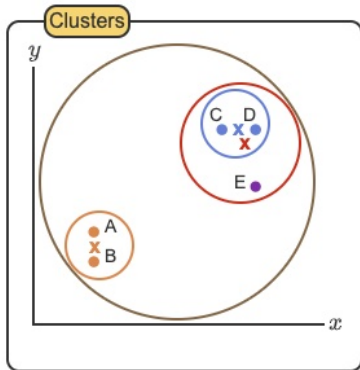
Dendrograms



Dendrograms



Dendrograms



The Algorithm

- 1) Start with all n observations and a measure (Euclidean distance for example) of pairwise dissimilarities. Treat each observation as its own cluster;
- 2) For $i = n, n - 1, \dots, 2$:
 - (a) Examine all pairwise inter-cluster dissimilarities among the i clusters and identify the pair of clusters that are most similar. Fuse these two clusters.
 - (b) Compute the new pairwise inter-cluster dissimilarities among the $i - 1$ remaining clusters.

Clustering: practical issues

- Scaling of variables matters!
- In the case of hierarchical clustering:
 - ▶ What metric (dissimilarity measure) should be used? So far, we have used the Euclidean distance as the dissimilarity measure, but in some cases it might be more useful to use something else (for eg. a *correlation-based distance*).
 - ▶ What kind of linkage should be used?
- How many clusters to choose? (in both K -Means clustering or hierarchical clustering).
- Which features should we use to drive the clustering?